Retreat Plan

Topic I: ALM

Shrink the Dual Regularization, Keep the Penalty: Tight Complexity of Linearized Augmented Lagrangian Methods

Target Journal

SIAM Journal on Optimization (SIOPT)

听众

对象	Relevance
Axel Böhm Postdoc, Univ. Vienna	**Focus**: weakly convex composite optimization, variable smoothing, nonconvex-concave minimax **Link**: variable smoothing 对“光滑函数 + 弱凸函数复合线性算子”给出 $\mathcal{O}(\epsilon^{-3})$ [1]。本文 $\beta = 0$ 时，对 λ 取最大后的目标是 f 加 $h \circ \mathcal{A}$ 的 Moreau 包络（参数 $1/(\rho + s)$）；$s = \Theta(1/\epsilon)$ 对应光滑参数 $\Theta(\epsilon)$，与 [1] 的设置相同。下界实例的 $\mathcal{A}$ 为仿射映射，落在 [1] 的问题类中。交替近端 GDA [2] 是最接近的同类方法。 [1] A. Böhm and S. J. Wright. Variable smoothing for weakly convex composite functions. J. Optim. Theory Appl., 2021. [2] R. I. Boţ and A. Böhm. Alternating proximal-gradient steps for (stochastic) nonconvex-concave minimax problems. SIAM J. Optim., 2023.
Radu Ioan Boţ Professor, Univ. Vienna	**Focus**: Tikhonov regularization, anchoring, monotone operator flows **Link**: flexible anchoring extragradient 由 Tikhonov 正则单调流离散得到，anchor 固定，权重 $\varepsilon^k \to 0$ [3]。本文 $\beta = 0$ 时，对偶中心保持为零，对偶正则项 $-\frac{1}{2s}\|\lambda\|^2$ 是以零为 anchor、权重 $1/s$ 固定的 Tikhonov 正则；对 SIOPT 定位的判断 [3] R. I. Boţ and E. Chenchene. Extra-gradient method with flexible anchoring: strong convergence and fast residual decay. SIAM J. Optim., to appear; arXiv:2410.14369.
Ernö Robert Csetnek Senior researcher, Univ. Vienna; COAP editorial board	**Focus**: fast OGDA, Tikhonov-regularized dynamical systems, strong convergence **Link**: $\mathcal{C}_\theta$ 只含单步梯度下降–上升更新，对偶步加入外推或 optimistic 修正 [4] 后的能否绕过下界 [4] R. I. Boţ, E. R. Csetnek, and D.-K. Nguyen. Fast optimistic gradient descent ascent (OGDA) method in continuous and discrete time. Found. Comput. Math., 2025.
Yura Malitsky Assistant Professor, Univ. Vienna	**Focus**: primal-dual methods, adaptive stepsizes, variational inequalities **Link**: 线性约束下 PDHG 可写成纯 primal 方法 [5]；本文的组合罚参数恒等式在 $\mathcal{A}$ 线性时与这一 primal 改写同类；自适应步长能否绕过下界 [6] [5] Y. Malitsky. The primal-dual hybrid gradient method reduces to a primal method for linearly constrained optimization problems. arXiv:1706.02602, 2017. [6] Y. Malitsky and K. Mishchenko. Adaptive proximal gradient method for convex optimization. NeurIPS, 2024.